3) The derivative of a function $f = \sin x$ is zero at the points $x = \pm \pi n$ (at these points the tangent to the graph of the function is horizontal).

This "list" could, of course, be expanded.

Next I would like to attract your attention to the following: from the viewpoint of mathematics a derivative of a function must also be considered as a certain $\boxed{\text{function}}$.

$\boxed{\text{Reader:}}$ But the derivative is a limit, and, consequently, a $\boxed{\text{number}}$!

$\boxed{\text{Author:}}$ Let us clarify this. We have fixed a point $x = x_0$ and obtained for a function $f(x)$ at this point the number $\boxed{\lim_{\Delta x \to 0} \frac{\Delta f(x_0)}{\Delta x}}$.

For each point x (from the domain of f) we have, in the general case, its own number $\boxed{\lim_{\Delta x \to 0} \frac{\Delta f(x)}{\Delta x}}$

This gives a mapping of a certain set of numbers x onto a different set of numbers $\lim_{\Delta x \to 0} \frac{\Delta f(x)}{\Delta x}$. The function which represents this mapping of one numerical set onto another is said to be the derivative and is denoted by $f'(x)$.

$\boxed{\text{Reader:}}$ I see. So far we have considered $\boxed{\text{only one value}}$ of the function $f'(x)$, namely, its value at the point $x = x_0$.

$\boxed{\text{Author:}}$ I would like to remind you that in the previous dialogue we analyzed $v(t)$ which was the derivative of $s(t)$. The graphs of the two functions (i.e. the function $s(t)$ and its derivative $v(t)$) were even compared in Fig. 35.

$\boxed{\text{Reader:}}$ Now it is clear.

$\boxed{\text{Author:}}$ I would like to make two remarks with regards to $f'(x)$.

$\boxed{\text{Note I:}}$ A function $f'(x)$ is obtained only by using a function $f(x)$.

Indeed, $$\boxed{f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}} \quad (4)$$